Experiment No.: 12 Name: Vansh Mahant PRN: 24070521038 Section: A Batch: A1

1. Title

Library Book Tracker: DOM manipulation, event handling, Web Storage (localStorage / sessionStorage) and jQuery AJAX.

2. Software / Tools Used

• HTML5, CSS3 and JavaScript

• jQuery 3.7.1 (CDN) – $.ajax(), $.each(), $.grep(), event delegation

• Web Storage: localStorage and sessionStorage; JSON file (books.json)

• Visual Studio Code, Web Browser, local web server

3. Program Code

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Library Book Tracker</title>
  <link rel="stylesheet" href="style.css">
  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
</head>
<body>
  <main class="app">
    <header>
      <h1>Library Book Tracker</h1>
      <p>DOM manipulation, events, Web Storage and jQuery AJAX</p>
    </header>

    <!-- Status bar -->
    <section class="status">
      <div>AJAX status: <b id="ajaxStatus">Not requested</b></div>
      <div>Last request: <b id="ajaxTime">-</b></div>
      <div>Saved in: <b>localStorage</b> (books) &amp; <b>sessionStorage</b> (search)</div>
    </section>
    <div id="msg" class="msg" role="status"></div>

    <!-- Add book form -->
    <section class="card">
      <h2>Add a Book</h2>
      <form id="bookForm" novalidate>
        <div class="fields">
          <div><label for="title">Title</label><input id="title" type="text" placeholder="Book
title"></div>
          <div><label for="author">Author</label><input id="author" type="text" placeholder="Author
name"></div>
          <div><label for="genre">Genre</label>
            <select id="genre">
              <option>Programming</option><option>Self-Help</option><option>Biography</option>
              <option>Fiction</option><option>Science</option>
            </select></div>
          <div><label for="year">Year</label><input id="year" type="number" placeholder="e.g.
2021"></div>
        </div>
        <div class="err" id="formErr"></div>
        <div class="buttons">
          <button type="submit">Add Book</button>
          <button type="button" id="loadBtn" class="green">Load from Server (AJAX)</button>
          <button type="button" id="badBtn" class="amber">Test AJAX Error</button>
          <button type="button" id="clearBtn" class="red">Clear All</button>
```

```
        </div>
      </form>
    </section>

    <!-- Search + filter -->
    <section class="card">
      <div class="tools">
        <input id="search" type="search" placeholder="Search by title or author...">
        <select id="filterGenre"><option value="All">All genres</option></select>
        <select id="filterRead">
          <option value="all">All books</option>
          <option value="read">Read</option>
          <option value="unread">Unread</option>
        </select>
      </div>

      <table>
        <thead>
          <tr><th>ID</th><th>Title</th><th>Author</th><th>Genre</th><th>Year</th><th>Status</
th><th>Actions</th></tr>
        </thead>
        <tbody id="bookBody"></tbody>
      </table>

      <div class="stats">
        <span>Total: <b id="sTotal">0</b></span>
        <span>Read: <b id="sRead">0</b></span>
        <span>Unread: <b id="sUnread">0</b></span>
        <span>Progress: <b id="sPct">0%</b></span>
      </div>
    </section>

    <footer>Vansh Mahant | PRN: 24070521038 | Section A | Batch A1</footer>
  </main>

  <script src="app.js"></script>
</body>
</html>
```

style.css

```
:root { --ink:#1e293b; --muted:#64748b; --line:#e2e8f0; --brand:#4338ca; --brand-d:#3730a3; --
bg:#f1f5f9; }
* { box-sizing:border-box; }
body { margin:0; font-family:"Segoe UI", Arial, sans-serif; background:var(--bg); color:var(--
ink); }
.app { max-width:980px; margin:0 auto; padding:26px 16px 18px; }
header { text-align:center; margin-bottom:16px; }
header h1 { margin:0 0 4px; color:var(--brand-d); }
header p { margin:0; color:var(--muted); font-size:14px; }
.status { display:flex; flex-wrap:wrap; justify-content:space-between; gap:8px; background:#fff;
border:1px solid var(--line); border-radius:10px; padding:10px 16px; font-size:13px; color:var(--
muted); margin-bottom:12px; }
.status b { color:var(--brand-d); }
.msg { display:none; padding:10px 14px; border-radius:8px; font-size:14px; margin-bottom:12px; text-
align:center; }
.msg.ok { background:#dcfce7; color:#166534; } .msg.info { background:#e0e7ff;
color:#3730a3; } .msg.bad { background:#fee2e2; color:#991b1b; }
.card { background:#fff; border:1px solid var(--line); border-radius:12px; padding:18px; margin-
bottom:16px; box-shadow:0 1px 3px rgba(0,0,0,.05); }
.card h2 { margin:0 0 10px; font-size:18px; color:var(--brand-d); }
.fields { display:grid; grid-template-columns:2fr 2fr 1.3fr 1fr; gap:12px; }
label { display:block; font-size:13px; font-weight:600; margin:0 0 4px; }
input, select { width:100%; padding:9px 10px; border:1px solid #cbd5e1; border-radius:6px; font-
size:14px; background:#fff; }
.err { color:#dc2626; font-size:13px; min-height:18px; margin-top:8px; }
.buttons { display:flex; flex-wrap:wrap; gap:8px; margin-top:6px; }
button { padding:9px 16px; border:0; border-radius:6px; background:var(--brand); color:#fff; font-
weight:600; cursor:pointer; }
button:hover { background:var(--brand-d); }
button.green { background:#15803d; } button.amber { background:#b45309; } button.red
{ background:#b91c1c; }
button.small { padding:4px 9px; font-size:12px; background:#475569; }
button.small.del { background:#dc2626; }
.tools { display:grid; grid-template-columns:2fr 1fr 1fr; gap:10px; margin-bottom:12px; }
table { width:100%; border-collapse:collapse; font-size:14px; }
th, td { padding:10px; border-bottom:1px solid var(--line); text-align:left; }
th { background:#eef2ff; color:var(--brand-d); }
tr.is-read td.t { color:#94a3b8; text-decoration:line-through; }
tr.empty td { text-align:center; color:var(--muted); font-style:italic; padding:20px; }
```

```
.pill { font-size:11px; font-weight:700; padding:3px 9px; border-radius:10px; }
.pill.yes { background:#dcfce7; color:#166534; } .pill.no { background:#fef3c7; color:#92400e; }
.stats { display:flex; justify-content:space-around; flex-wrap:wrap; gap:10px; margin-top:14px;
padding:12px; background:#f8fafc; border-radius:8px; font-size:14px; }
footer { text-align:center; color:var(--muted); font-size:13px; }
@media (max-width:760px){ .fields,.tools{grid-template-columns:1fr 1fr;} }
@media (max-width:480px){ .fields,.tools{grid-template-columns:1fr;} }
```

app.js

```
// Library Book Tracker - jQuery DOM, events, Web Storage and AJAX

const BOOKS_KEY = "libraryBooks";      // localStorage key
const SEARCH_KEY = "lastSearch";       // sessionStorage key

// ---------- Web Storage helpers ----------
function loadBooks() {
  try {
    return JSON.parse(localStorage.getItem(BOOKS_KEY)) || [];
  } catch (e) {
    return [];
  }
}
function saveBooks(list) {
  localStorage.setItem(BOOKS_KEY, JSON.stringify(list));
}

// ---------- small UI helpers ----------
let msgTimer;
function showMsg(text, kind) {
  clearTimeout(msgTimer);
  $("#msg").removeClass("ok info bad").addClass(kind).text(text).show();
  msgTimer = setTimeout(function () { $("#msg").fadeOut(250); }, 3500);
}

function escapeHtml(s) {
  return String(s).replace(/[&<>"']/g, function (c) {
    return { "&": "&amp;", "<": "&lt;", ">": "&gt;", '"': "&quot;", "'": "&#39;" }[c];
  });
}

function nextId(list) {
  let max = 100;
  $.each(list, function (i, b) {
    const n = parseInt(String(b.id).replace("B-", ""), 10);
    if (n > max) max = n;
  });
  return "B-" + (max + 1);
}

// ---------- render the table ----------
function render() {
  const books = loadBooks();
  const term = $("#search").val().trim().toLowerCase();
  const genre = $("#filterGenre").val();
  const readFilter = $("#filterRead").val();

  // rebuild the genre dropdown from the data (keep current choice)
  const genres = [];
  $.each(books, function (i, b) { if ($.inArray(b.genre, genres) === -1) genres.push(b.genre); });
  const $g = $("#filterGenre");
  const keep = $g.val() || "All";
  $g.empty().append('<option value="All">All genres</option>');
  $.each(genres.sort(), function (i, g) { $g.append($("<option>").val(g).text(g)); });
  $g.val(genres.indexOf(keep) !== -1 ? keep : "All");

  const shown = $.grep(books, function (b) {
    const matchText = (b.title + " " + b.author).toLowerCase().indexOf(term) !== -1;
    const matchGenre = $g.val() === "All" || b.genre === $g.val();
    const matchRead = readFilter === "all" || (readFilter === "read" ? b.read : !b.read);
    return matchText && matchGenre && matchRead;
  });

  const $body = $("#bookBody").empty();
  if (shown.length === 0) {
    $body.append('<tr class="empty"><td colspan="7">No books to show. Add one or load from the
server.</td></tr>');
  }
  $.each(shown, function (i, b) {
    $body.append(
```

```
      '<tr data-id="' + b.id + '" class="' + (b.read ? "is-read" : "") + '">' +
        "<td>" + escapeHtml(b.id) + "</td>" +
        '<td class="t">' + escapeHtml(b.title) + "</td>" +
        "<td>" + escapeHtml(b.author) + "</td>" +
        "<td>" + escapeHtml(b.genre) + "</td>" +
        "<td>" + escapeHtml(b.year) + "</td>" +
        '<td><span class="pill ' + (b.read ? "yes" : "no") + '">' + (b.read ? "Read" : "Unread") +
"</span></td>" +
        '<td><button class="small toggle">' + (b.read ? "Mark Unread" : "Mark Read") + "</button> "
+
        '<button class="small del">Delete</button></td>' +
      "</tr>"
    );
  });

  const total = books.length;
  const read = $.grep(books, function (b) { return b.read; }).length;
  $("#sTotal").text(total);
  $("#sRead").text(read);
  $("#sUnread").text(total - read);
  $("#sPct").text(total ? Math.round(read * 100 / total) + "%" : "0%");
}

// ---------- AJAX ----------
function loadFromServer(url) {
  $("#ajaxStatus").text("Requesting " + url + " ...");
  $.ajax({
    url: url,
    type: "GET",
    dataType: "json",
    cache: false
  })
    .done(function (data) {
      const books = loadBooks();
      let added = 0;
      $.each(data, function (i, item) {
        const exists = $.grep(books, function (b) { return b.id === item.id; }).length > 0;
        if (!exists) { books.push(item); added++; }
      });
      saveBooks(books);
      render();
      $("#ajaxStatus").text("Success (HTTP 200)");
      showMsg(data.length + " records received, " + added + " new book(s) added.", "ok");
    })
    .fail(function (xhr, status, error) {
      $("#ajaxStatus").text("Failed (" + (xhr.status || "no response") + ")");
      showMsg("Could not load " + url + ". Run the page from a local server.", "bad");
    })
    .always(function () {
      $("#ajaxTime").text(new Date().toLocaleTimeString());
    });
}

// ---------- events ----------
$(document).ready(function () {
  // restore the last search typed in this browser tab (sessionStorage)
  $("#search").val(sessionStorage.getItem(SEARCH_KEY) || "");
  render();

  // submit : add a book
  $("#bookForm").on("submit", function (e) {
    e.preventDefault();
    const title = $("#title").val().trim();
    const author = $("#author").val().trim();
    const year = parseInt($("#year").val(), 10);
    const thisYear = new Date().getFullYear();

    if (title === "" || author === "") {
      $("#formErr").text("Title and author are required.");
      return;
    }
    if (isNaN(year) || year < 1000 || year > thisYear) {
      $("#formErr").text("Enter a valid year between 1000 and " + thisYear + ".");
      return;
    }
    $("#formErr").text("");

    const books = loadBooks();
```

```
    books.unshift({ id: nextId(books), title: title, author: author, genre: $("#genre").val(), year:
year, read: false });
    saveBooks(books);
    render();
    this.reset();
    showMsg('"' + title + '" added and saved to localStorage.', "ok");
  });

  // click : AJAX buttons
  $("#loadBtn").on("click", function () { loadFromServer("books.json"); });
  $("#badBtn").on("click", function () { loadFromServer("missing-file.json"); });

  // click : clear everything (with confirm)
  $("#clearBtn").on("click", function () {
    if (confirm("Remove all books from localStorage?")) {
      localStorage.removeItem(BOOKS_KEY);
      render();
      showMsg("All books removed.", "info");
    }
  });

  // keyup : live search (also remembered in sessionStorage)
  $("#search").on("keyup input", function () {
    sessionStorage.setItem(SEARCH_KEY, $(this).val());
    render();
  });

  // change : filters
  $("#filterGenre, #filterRead").on("change", render);

  // event delegation : buttons inside rows that are created dynamically
  $("#bookBody").on("click", ".toggle", function () {
    const id = $(this).closest("tr").data("id");
    const books = loadBooks();
    $.each(books, function (i, b) { if (b.id === id) b.read = !b.read; });
    saveBooks(books);
    render();
  });

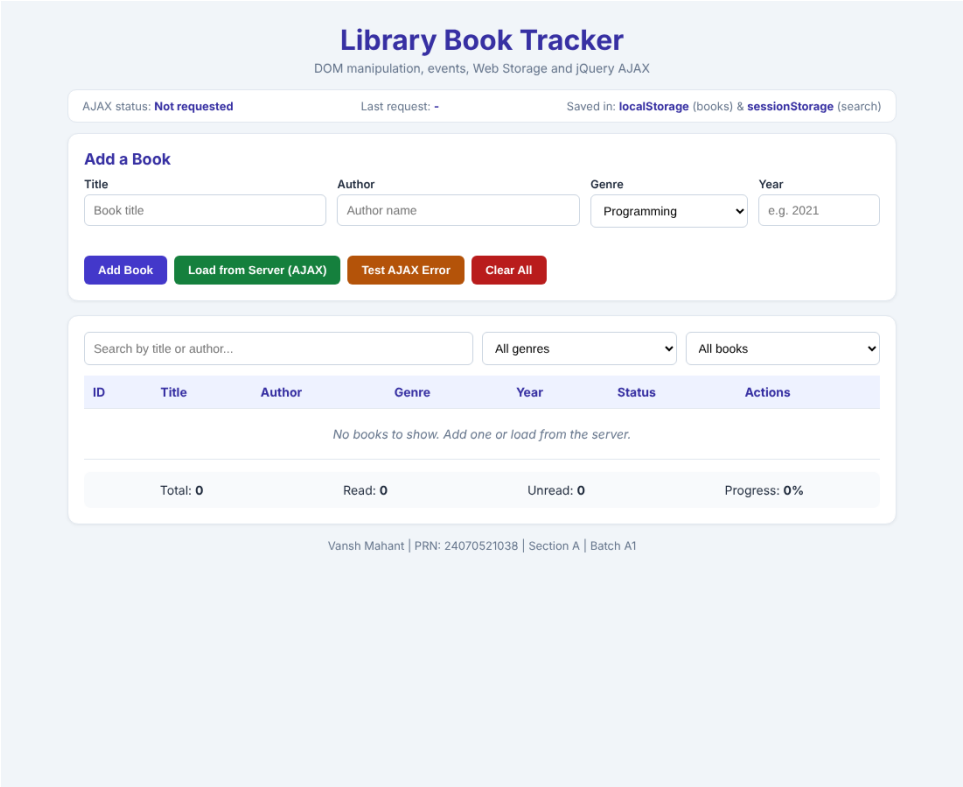
  $("#bookBody").on("click", ".del", function () {
    const id = $(this).closest("tr").data("id");
    saveBooks($.grep(loadBooks(), function (b) { return b.id !== id; }));
    render();
    showMsg("Book " + id + " deleted.", "info");
  });
});
```

books.json

```
[
  { "id": "B-101", "title": "Eloquent JavaScript", "author": "Marijn Haverbeke", "genre":
"Programming", "year": 2018, "read": true },
  { "id": "B-102", "title": "You Don't Know JS Yet", "author": "Kyle Simpson", "genre":
"Programming", "year": 2020, "read": false },
  { "id": "B-103", "title": "Atomic Habits", "author": "James Clear", "genre": "Self-Help", "year":
2018, "read": true },
  { "id": "B-104", "title": "The Pragmatic Programmer", "author": "Andrew Hunt", "genre":
"Programming", "year": 1999, "read": false },
  { "id": "B-105", "title": "Wings of Fire", "author": "A. P. J. Abdul Kalam", "genre": "Biography",
"year": 1999, "read": true }
]
```

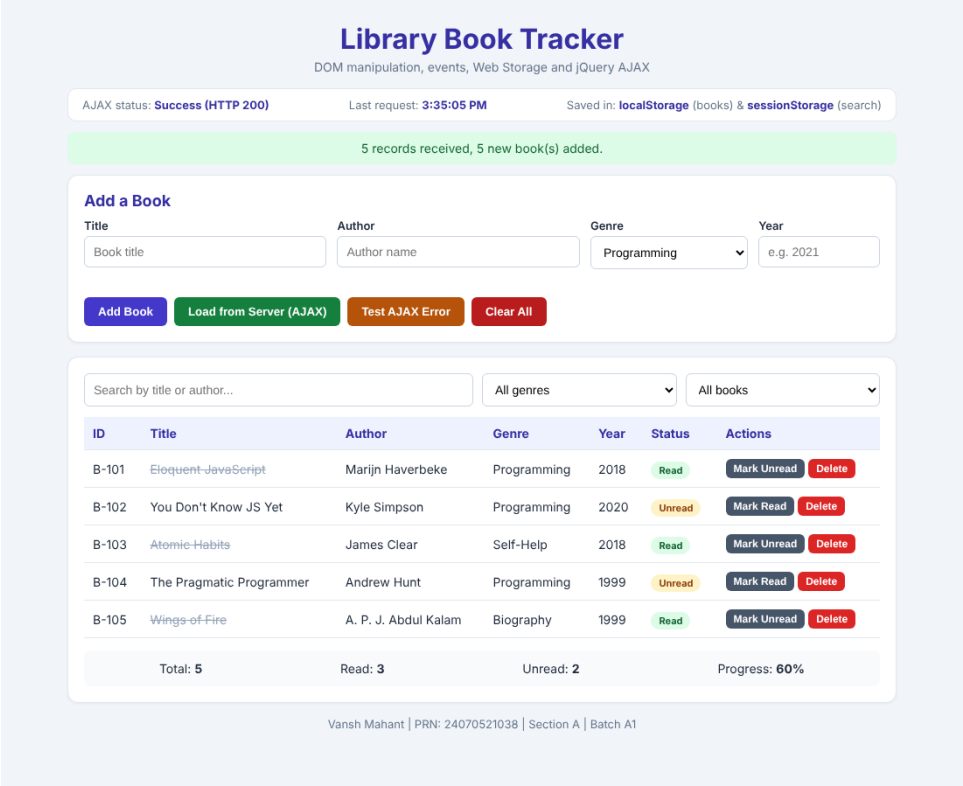
4. Output

Initial page with the Add Book form, search and filters, and AJAX status bar.



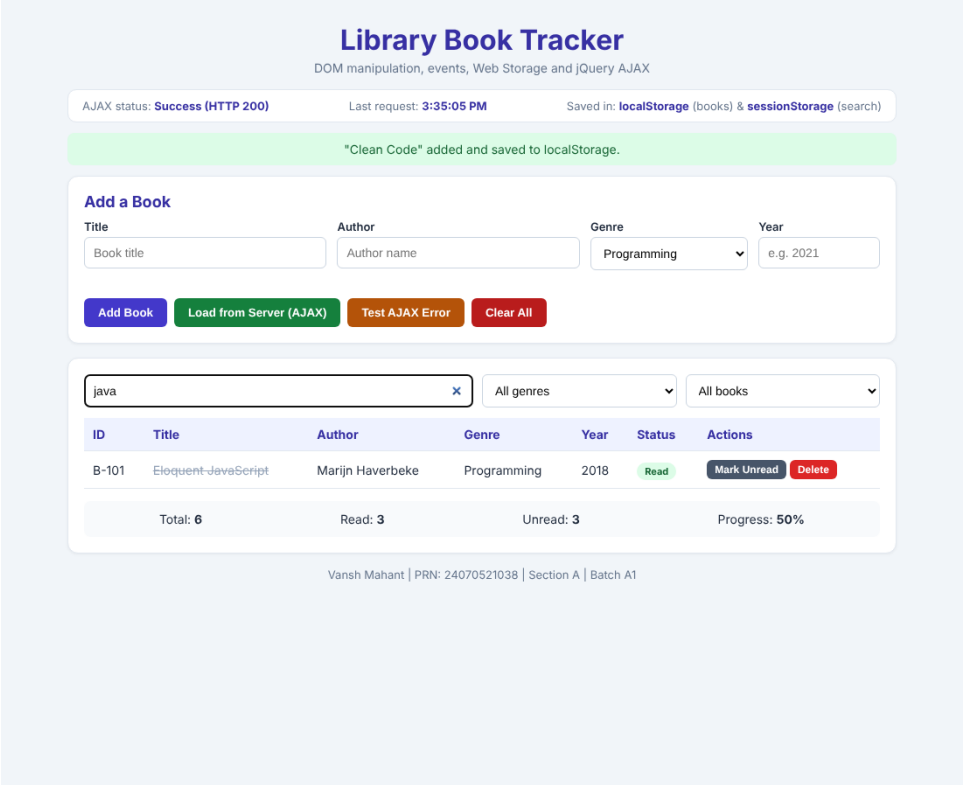
PRN: 24070521038 | Name: Vansh Mahant | File: index.html

After 'Load from Server (AJAX)': books.json loaded, status Success (HTTP 200).



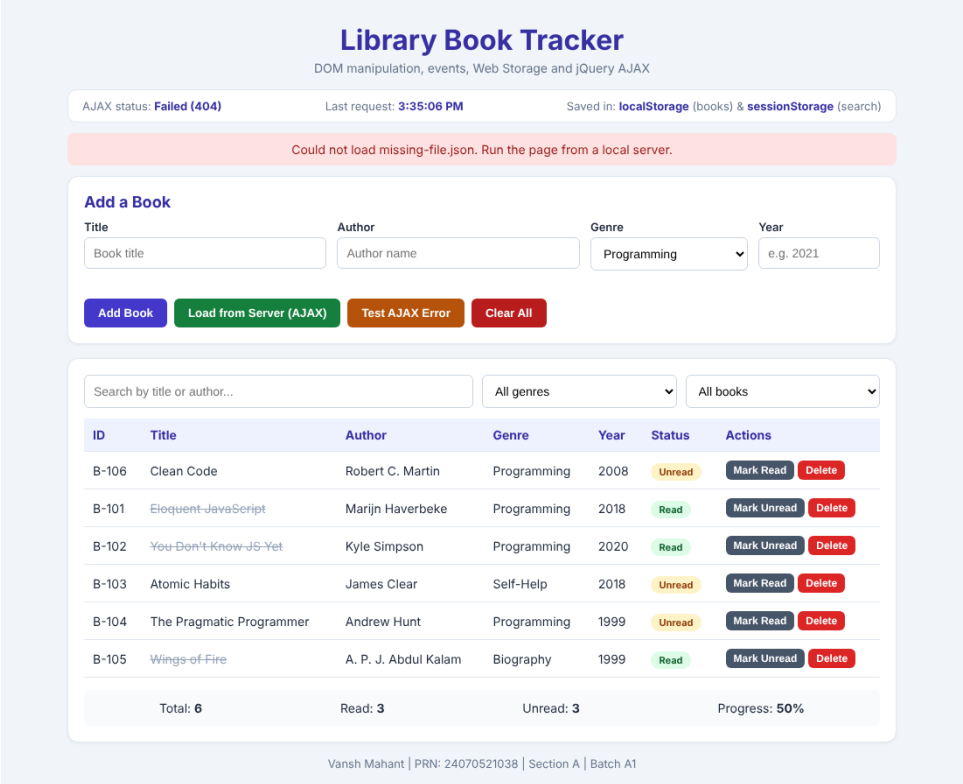
PRN: 24070521038 | Name: Vansh Mahant | File: index.html

New book added and saved to localStorage; books marked Read; live search for 'java' (stored in sessionStorage).



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'Test AJAX Error': request for a missing file handled by .fail() (HTTP 404 message).



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Code must also be uploaded to GitHub.
Practical 12_24070521038_Vansh Mahant_A_A1

5. Result / Conclusion

DOM elements were created and updated with jQuery. Events (submit, click, keyup, change) and event delegation were used for dynamic rows. Books are stored in localStorage and the last search in sessionStorage. Data was fetched from books.json with $.ajax(), with success and error handling.